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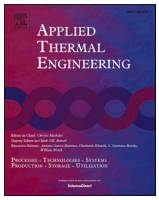

Research Paper

Numerical analysis on the cooling and electrical performances of photovoltaic noise barrier

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ABSTRACT

A photovoltaic noise barrier (PVNB) is a useful device for installing PV systems on a noise barrier, which can save the space and cost associated with the installation of PV systems. Since a noise barrier usually consists of porous materials with low thermal conductivities, investigation on the cooling performance of the PVNBs is essential because the temperature of the PV module has a significant impact on its electrical performance. In this study, the cooling and electrical performances of a PVNB were numerically investigated using four different models: PV module-only, PVNB model without a heat sink, PVNB model with a conventional heat sink, and PVNB model with the proposed heat sink configuration. The simulation was validated by comparing the predicted and measured cell temperatures of the PV module at different heat fluxes, and the results showed good agreement. The average cell temperature in the PVNB model without a heat sink significantly increased by 42.2% from 86.98 ℃ to 123.66 ℃ because of the low thermal conductivity of the noise absorber, compared to the PV module only case. In the case of proposed heat sink configuration, the average cell temperature decreased by 42.7% from 123.66 ℃ to 70.90 ℃, compared with the PVNB model without a heat sink, owing to the additional injection of the cold ambient temperature air to the heat sink channel as well as natural convection and stack effect. The electrical performance of the PVs in the tested PVNBs was analyzed, and the results indicated that the maximum output power and voltage in the PVNB with the proposed heat sink configuration increased by 26.6% and 19.0%, respectively, compared to the PVNB model without a heat sink. The results of this study provide useful fundamentals regarding natural convection and the stack effect for optimizing heat sink designs for PVNB.

1. Introduction

Photovoltaic (PV) systems have been utilized as next-generation power sources because they directly convert solar energy into electric energy [1] and emit little greenhouse gas [2]. Even though PV systems are truly a technology to counteract the energy crisis and global warming, the fact that PV systems require considerable installation areas can be a fatal disadvantage, especially for large-scale power generation [3]. Thus, seeking a method to combine PV systems with existing infrastructure could be an attractive solution to this issue. One example is the photovoltaic noise barrier (PVNB), which acts as both a noise barrier and a PV system by installing a PV module on noise barriers that can be installed along roads or railways. PVNBs were first introduced in Switzerland in 1989 [4], and numerous countries have installed them or noticed their benefits after their success in Switzerland.

Several studies have been conducted on the power generation performance and effects of using PVNBs. Wadhawan et al. [5] estimated the energy potential using a mathematical model under the assumption that PV modules are installed along existing noise barriers in the U.S. They reported that the power potential ranged from 7 to 9 GW, which is sufficient to power more than 50,000 houses in the U.S. Gu et al. [6] estimated the energy potential and environmental effect of using PVNBs along a metro line in China and reported that they can generate approximately 5000 kWh per year and significantly reduce SO_2 and CO_2 emissions. Faturrochman et al. [7] investigated the effects of orientation, tilting angle, location, cell position, and bypass diode configuration on the annual energy yield of bifacial PVNBs.

Cooling of the PV system is important because the efficiency decreases as the temperature of the PV module increases. For a typical

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Nomenclature

A	area (m^2)
c_p	specific heat at constant pressure (J kg^{-1} K^{-1})
h	heat transfer coefficient (W m^{-2} K^{-1})
I	current (A)
k	thermal conductivity (W m^{-1} K^{-1})
k_B	Boltzmann constant (–)
$\dot{m}$	mass flow rate (kg s^{-1})
N	diode quality factor (–)
P	electric power (W)
P'	modified pressure (Pa)
q	electronic charge (C)
Q	heat flux (kW m^{-2})
R	electric resistance (Ω)
T	temperature (℃)
U	velocity (m/s)
V	voltage (V)
x	length from the center of the module (mm)

Greek symbols

α	diode quality factor (–)
β	temperature coefficient of V_{oc} (–)
ε	stress tensor (–)
θ	grid-dependency index (%)
μ	dynamic viscosity (Pa s)
μ_{eff}	effective viscosity (Pa s)
ρ	density (kg m^{-3})
ρ_0	density of reference ambient air (kg m^{-3})
τ	shear stress (Pa)
ψ_{ij}	Kronecker delta function (–)

Subscripts

avg	averaged
c	channel
flu	fluid
m	mean
max	maximum
min	minimum
oc	open circuit
ocref	open circuit at reference temperature
ph	photogenerated
ref	reference temperature
rs	reverse saturation
s	series
sc	short circuit
sh	shunt
sol	solid

Acronyms

MAE	mean absolute error
MPP	maximum power point
NOCT	nominal operating cell temperature
PERC	mean absolute relative deviation (%)
PV	photovoltaic
PVNB	photovoltaic noise barrier

monocrystalline-silicon PV system, every 1 ℃ increase in the temperature decreases the efficiency of the system by approximately 0.45% [8]. Typical cooling methods for PV systems can be classified into two groups: passive and active cooling. Bayrak et al. [9], Ahmad et al. [10], and Mankani et al. [11] investigated the optimization of passive cooling of PV modules using heat sinks, reporting that the cooling performance can be improved by changing the parameters of the heat sinks, such as the spacing, height, thickness, and fin material. Elminshawy et al. [12] investigated an active cooling system for PV modules using an air pre-cooled by the geothermal heat exchanger. They reported that the pre-cooled air by the geothermal heat exchanger decreased the PV module temperature by 13 ℃, resulting in an 18.9% of increase in the power output. Čabo et al. [13] studied a passive PV module cooling system with aluminum fins attached to the backside of the PV module. Their system showed a 5% of increase in the power output owing to the decrease in the PV module temperature with an economical price with a payback period of 5 years. Selimefendigil and Öztop [14] numerically analyzed the cooling performance of fins with various geometry. They concluded that the channel with porous layer and wavy walls had the highest cooling performance with 29–37.6 ℃ decrease in the PV module temperature. Hernandez-Perez et al. [15] proposed a heat sink with discontinuous segments and studied the thermal performance of the proposed heat sink, reporting that the proposed heat sink decreased the PV module temperature by 7 ℃ compared to the conventional heat sink. Wang et al. [16] investigated the energy-saving performance of a water-cooled PV system when the excess cooling capacity generated by the chiller was stored. Lillo-Bravo et al. [17] compared the influence of the level of architectural PV integration on the PV energy balance between a natural ventilation cooling system and forced cooling system.

The temperature issue of PV modules is much more significant when combined with noise barriers. Because noise barrier materials are usually porous media, such as glass wool, which can absorb sound effectively, the thermal conductivities of the noise barriers are substantially low owing to the air trapped inside the porous media. Therefore, the heat exchange between the outside air or coolant and the PV module is less active for the PVNB than for the PV module alone. Although existing studies on PVNB have investigated the advantages and possible power generation of PVNBs, studies on cooling technologies for PVNBs to suppress performance degradation are limited in the existing literature. Cooling performance of the PVNB is crucial because the amount of power generation is directly affected by the module temperature and the PV operates over a year for its entire lifespan of around 30–35 years.

This study investigated the necessity of cooling in PVNBs and the cooling performance of a conventional heat sink in PVNBs by analyzing the flow and heat transfer characteristics of natural convection. In addition, a heat sink with a better cooling performance of the PVNB than the conventional heat sink was proposed by changing the flow configuration. A passive cooling method using natural convection was employed owing to the practical maintenance and the absence of additional power input or coolant; PVNBs are aimed to be installed mostly in large-scale construction. The numerical simulation was validated experimentally using measured data, and showed good agreement between the measured and predicted data. Finally, the electrical performance of the PVNBs was investigated to study the effects of cooling performance on the electrical performance based on the results of the numerical analysis. The results of this study can be used to understand the necessity of cooling in PVNBs by analyzing the degradation of cooling and electrical performance when a noise barrier is added to the PV module.

2. Simulation model

2.1. Simulation model and boundary conditions

In this study, four simulation models were used: the PV module-only model, the PVNB model without a heat sink, the PVNB model with a conventional heat sink, and the PVNB model with the proposed heat sink configuration. Schematic views of the four simulation models are shown

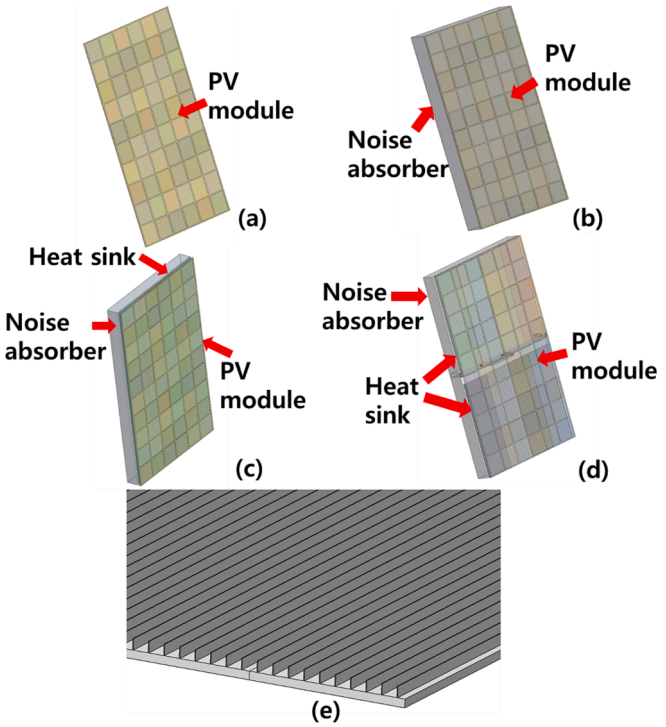


Fig. 1. Schematic view of (a) the PV module-only model, (b) PVNB model without a heatsink, (c) PVNB model with conventional heat sink, (d) PVNB model with the proposed heat sink configuration, and (e) the heat sink.

Table 1
Materials and dimensions of each solid domain.

Domain	Material	Dimensions (width × height × thickness)
PV module	Glass, Silicon, EVA	1000 mm × 1600 mm × 5 mm
Heat sink	Aluminum	1000 mm × 1600 mm × 30 mm
Noise absorber	Glass wool	1000 mm × 1600 mm × 100 mm

in Fig. 1(a)–(d). The simulation models comprised a combination of three solid domains: a PV module, a heat sink, and a noise absorber. A commercial monocrystalline PV module is used for the simulation. Generally, PV modules consist of several thin layers of different materials [18]. Therefore, the PV module was simplified into three layers, as shown in Fig. 1(a), considering the number of meshes and solving time. For the heat sink, a conventional simple heat sink with general geometry was used, considering practical maintenance and reasonable price; these two factors are the most important for PVNBs, which are generally installed as large-scale constructions. Therefore, a passive cooling method using natural convection was used in this study because it is preferable to an active cooling method. The heat sink was composed of aluminum fins with 23.8 mm distance from each other. A glass-wool noise absorber was used in this study. The materials and dimensions of each solid domain are listed in Table 1.

Fig. 2(a) shows the configuration of the PV module, and Fig. 2(b) presents a schematic view of the boundary conditions. The solar irradiance was set at 800 W m^{-2} and the ambient air temperature was set at 25 ℃, which is the conditions with clear skies and strong solar radiation is irradiated at a right angle to the module. According to the PV module manufacturer, the nominal output power was 224 W, and the reflectance of common PV modules was 5% [19]. Therefore, considering the output power and reflectance, the net heat flux of the PV module was set to 536 W m^{-2}. An open boundary was applied to the fluid domain at a relative pressure of 1 atm.

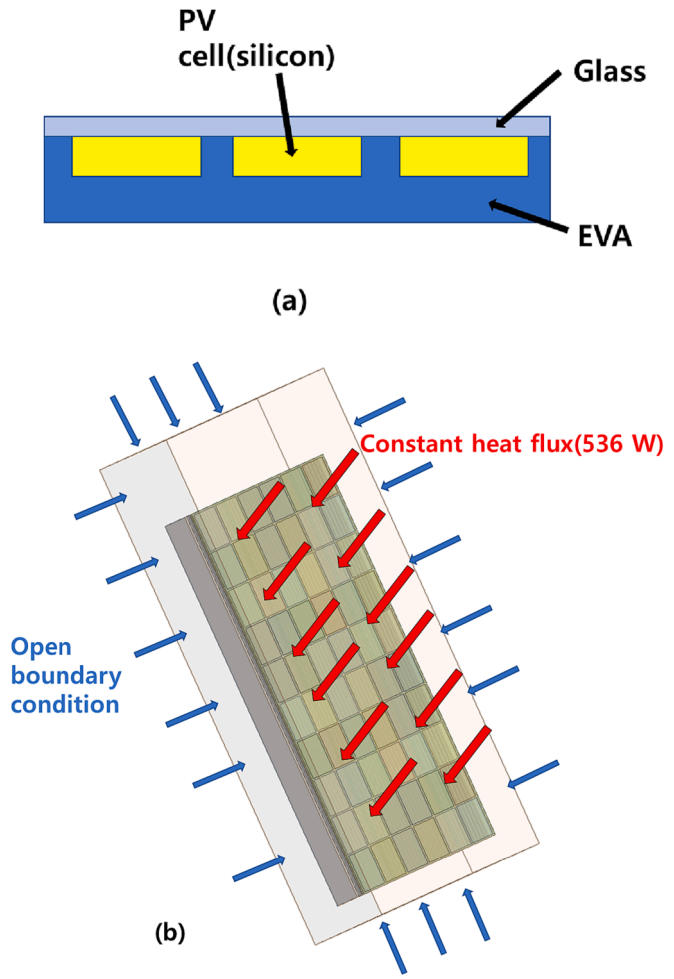


Fig. 2. (a) Simplification of the PV module domain. (b) Schematic view of the boundary conditions.

Table 2
Mechanical and thermal properties of the materials.

Material	Density(kg m^{-3})	Thermal conductivity (W/m ℃$^{-1}$)	Specific heat capacity(J kg^{-1} ℃$^{-1}$)
Glass	3000	6	800
Monocrystalline silicon	2300	158.7	710
EVA	960	10	2090
Aluminum	2702	237	903
Glass wool	48	0.034	670

2.2. Thermal-fluid model

The heat of the PV module was initially transferred via conduction to a heat sink, air, or a noise absorber. The conduction equation is as follows:

$$\frac{\partial}{\partial x_i}\left(k\frac{\partial T}{\partial x_i}\right) = Q \tag{1}$$

where Q denotes the heat flux radiating from the sun to the PV module. The thermal and mechanical properties of the materials are listed in Table 2.

Convective heat transfer occurs primarily in the fluid domain, and the heat of the PV module is transferred to the outside air via natural convection. The governing equations in the fluid domain are expressed

Table 3
Electrical properties and thermal coefficients of the commercial PV module.

Parameter	Unit	Value
Normal module operating temperature	℃	43 ± 3
Short circuit current (I_{sc})	A	7.92
Open circuit voltage (V_{oc})	V	37.31
Power at MPP (P_{mpp})	W	223.9
Temperature coefficient of V_{oc} (β)	% ℃$^{-1}$	0.28

as follows:

Continuity equation:

$$\frac{\partial \rho U_i}{\partial x_i} = 0 \tag{2}$$

Momentum equation:

$$\frac{\partial}{\partial x_i}(\rho U_i U_j) = -\frac{\partial p'}{\partial x_i} + \frac{\partial}{\partial x_j}\left[\mu_{eff}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right)\right] + (\rho - \rho_0) g_i \tag{3}$$

Energy equation:

$$\frac{\partial}{\partial x_i}(\rho c_p U_i T) = \frac{\partial}{\partial x_i}\left(k\frac{\partial T}{\partial x_i}\right) + \frac{\partial}{\partial x_i}(U_i(\tau_{ij})_{eff}) \tag{4}$$

The effective viscosity (μ_{eff}) and deviatoric stress tensor $(\tau_{ij})_{eff}$ are expressed by Eqs. (5) and (6), respectively:

$$\mu_{eff} = \mu\left(1 + \sqrt{\frac{C_\mu}{\mu}}\frac{k}{\sqrt{\varepsilon}}\right)^2 \tag{5}$$

$$(\tau_{ij})_{eff} = \mu_{eff}\left(\frac{\partial U_j}{\partial x_i} + \frac{\partial U_i}{\partial x_j}\right) - \frac{2}{3}\mu_{eff}\frac{\partial U_k}{\partial x_k}\psi_{ij} \tag{6}$$

Since some regions in the fluid domains were in turbulent flow range, especially for natural convection flow, the Menter shear stress transport turbulence model was used in the fluid domain. It is widely used because it provides a reliable prediction of the flow separation [20]. The transport equations for k and ω are expressed by Eqs. (7) and (8), respectively:

$$\frac{\partial}{\partial x_i}(\rho U_i \kappa) = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\kappa}\right)\frac{\partial \kappa}{\partial x_j}\right] + G_\kappa - Y_\kappa \tag{7}$$

$$\frac{\partial}{\partial x_i}(\rho U_i \omega) = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_t}{\sigma_\omega}\right)\frac{\partial \omega}{\partial x_j}\right] + G_\omega - Y_\omega + D_\omega \tag{8}$$

The following assumptions were used in the simulations: (1) the air was assumed to be an ideal gas, (2) radiative heat transfer was not considered, and (3) only natural convection was the source of air flow; other external forces that could affect air flow were not considered. The result of the preliminary simulation, which compared the simulations with and without radiative heat transfer, showed that the difference between the two simulations was trivial as the mean absolute error (MAE), which can be expressed by Eq. (9), was only 0.88%, meaning

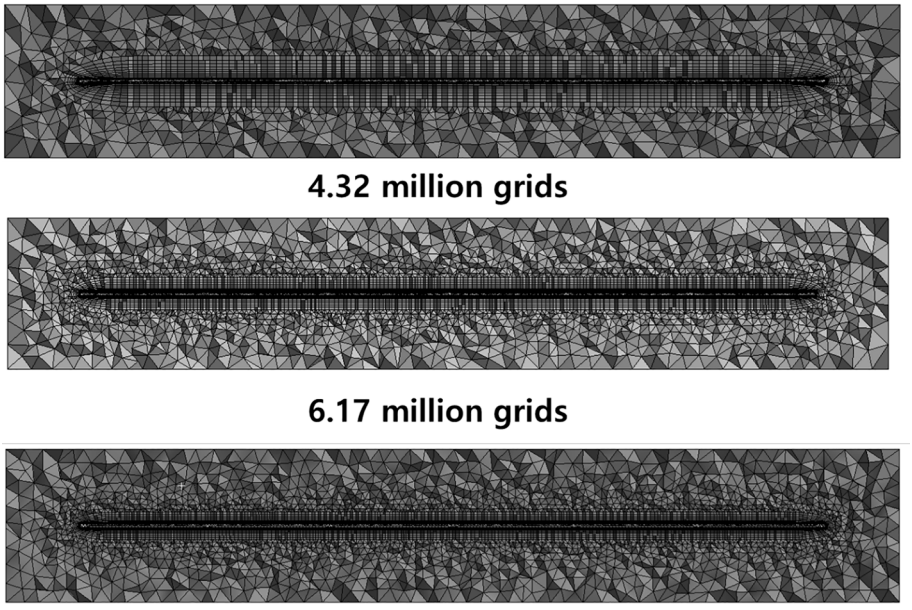


Fig. 3. Mesh characteristics of the PV module and fluid domain.

Table 4
Grid dependency test results for different heat flux values.

Heat flux	Numberof grids (Million)	T_{avg} (℃)	θ (%)	T_{max} (℃)	θ (%)	T_{min} (℃)	θ (%)	Remark
500 W	8.31	84.00	–	87.03	–	68.25	–	
	6.17	83.80	0.24	86.90	0.15	68.10	0.21	Accept
	4.32	85.34	1.84	88.66	2.02	69.80	2.50	
400 W	8.31	74.71	–	77.77	–	60.90	–	
	6.17	74.95	0.33	77.66	0.14	61.22	0.53	Accept
	4.32	75.94	1.32	79.22	2.01	62.35	1.84	
300 W	8.31	65.18	–	67.62	–	53.36	–	
	6.17	65.20	0.03	67.52	0.15	53.80	0.83	Accept
	4.32	65.78	0.89	68.53	1.50	54.61	1.51	

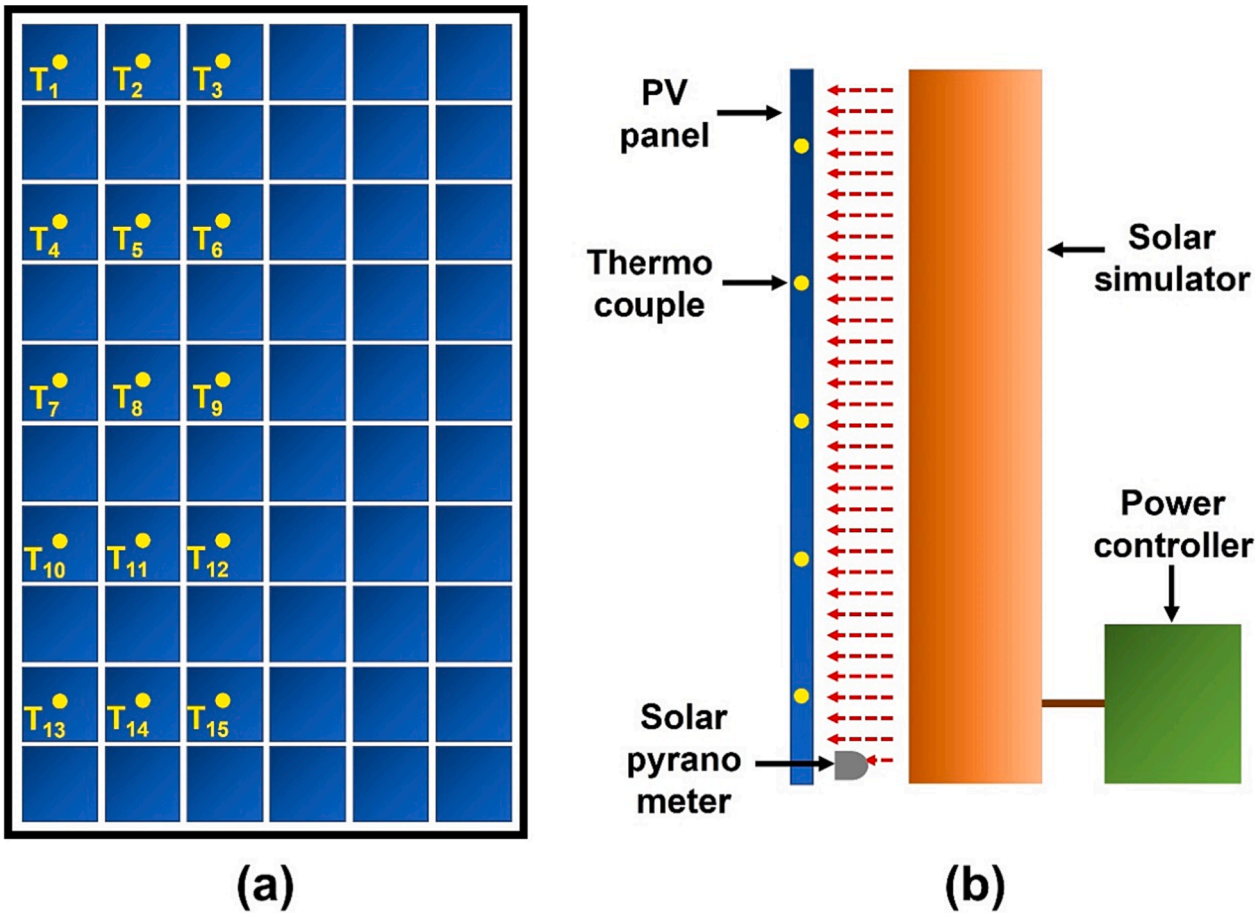


Fig. 4. Schematic of experiments: (a) locations of the thermocouples on the PV panel and (b) experimental setup.

that the radiative heat transfer can be neglected.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\frac{\left|T_{radiation\ considered} - T_{radiation\ neglected}\right|}{T_{radiation\ neglected}} \times 100 \tag{9}$$

2.3. Data reduction

The convective heat transfer coefficient was calculated using the temperatures of the solid surface (T_{sol}) and air (T_{flu}) obtained from the simulation, as shown in Eq. (10).

$$h = \frac{Q}{T_{sol} - T_{flu}} \tag{10}$$

where Q denotes wall heat flux. The outer surfaces of the PV and PVNB models were exposed to ambient air; therefore, T_{flu} was the same as the ambient temperature of 25 ℃. However, the air was surrounded by a heat sink inside the PVNB, resulting in typical internal flow in a square channel. Therefore, the mean temperature of the internal flow (T_m) was used to calculate T_{flu} inside the PVNB. T_m can be calculated as the mean temperature over the cross-sectional area of the channel, as shown in Eq. (11).

$$T_m = \frac{\int_{A_c}\left(\rho U c_p T\right)dA_c}{\dot{m}c_p} \tag{11}$$

The standard equation for the illuminated output current of a single PV cell is expressed as Eq. (12) [21].

$$I = I_{ph} - I_{rs}\left[\exp\left\{\left(\frac{q}{Nk_BT}\right)(V + IR_s)\right\} - 1\right] - \frac{(V + IR_s)}{R_{sh}} \tag{12}$$

The photogenerated current I_{ph} and the reverse saturation current I_{rs} are given by Eqs. (13) and (14) [22].

$$I_{ph} = \frac{BI_{sc} - \left(\frac{V_{oc}}{R_{sh}}\right)\Phi}{1 - \Phi} \tag{13}$$

$$I_{rs} = \frac{BI_{sc} - \left(\frac{V_{oc}}{R_{sh}}\right)}{\exp\left(\frac{qV_{oc}}{Nk_BT}\right)\left[1 - \exp\left\{\left(\frac{q}{Nk_BT}\right)\left(I_{sc}R_s - V_{oc}\right)\right\}\right]} \tag{14}$$

where

$$B = 1 + \left(\frac{R_s}{R_{sh}}\right) \tag{15}$$

$$\Phi = \frac{\exp\left(\frac{qI_{sc}R_s}{\alpha kT}\right) - 1}{\exp\left(\frac{qV_{oc}}{\alpha kT}\right) - 1} \tag{16}$$

The open-circuit voltage V_{oc} at the operating temperature is given by Eq. (17)

$$V_{oc} = V_{ocref}\left(1 - \frac{\beta}{100}\left(T_{ref} - T\right)\right) \tag{17}$$

Assuming that the PV cells are ideal and connected in series, the diode quality factor α is 1, the series resistance R_s is zero, and the shunt resistance R_{sh} has an infinite value [23]. Therefore, Eq. (12) can be simplified to Eq. (18).

$$I = I_{ph} - I_{rs}\left[\exp\left(\frac{qV}{k_BT}\right) - 1\right] \tag{18}$$

As the current changes significantly when the voltage changes from

Fig. 5. Pictures of experimental equipment.

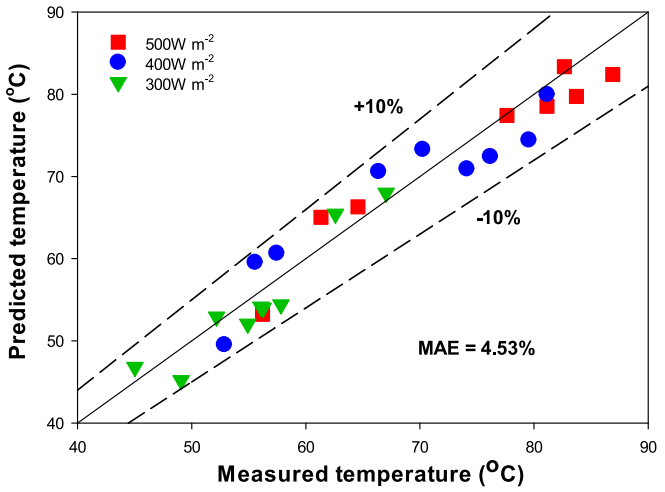


Fig. 6. Comparison of the measured and predicted temperatures.

the maximum power region to the open-circuit voltage region in typical *I–V* curves [24], it is preferable to express the load voltage of a single cell (V) in terms of the current (I), as shown in Eq. (19).

$$V = \frac{k_B T}{q} \ln\left(\frac{I_{ph} - I}{I_{rs}} + 1\right) \tag{19}$$

The performance indicator of the model is the maximum value of the summation of the PV output power. Because the mismatch effect of the cells is trivial, it can be neglected [25]. Therefore, the maximum output power can be expressed by Eq. (20).

$$P_{\max} = \left(I \sum_{i=1}^{n} V_i(I)\right)_{\max} \tag{20}$$

Properties of the commercial PV modules at the nominal operating cell temperature (NOCT) are listed in Table 3 [26].

2.4. Grid dependency test

Fig. 3 shows the generated mesh of the PV module and the fluid domain surrounding the PV module. Table 4 presents the grid-dependency test results for different heat flux values. The maximum and minimum cell temperatures and the average PV module temperature were calculated and used as parameters for the grid-dependency index, as given in Eq. (21) [27]. As a result of the grid-dependency test, 6.17 million grids were determined considering the computing power and accuracy, presenting a maximum grid-dependency index of 0.83% for the simulation in this study.

$$\theta = \frac{\left|i_{finer\ grid} - i_{coarser\ grid}\right|}{i_{finer\ grid}} \times 100 \tag{21}$$

3. Experiments for validation

Experiments were conducted to validate the simulation results by comparing the temperatures of PV cells. Fig. 4 shows a schematic of the experiments, including the locations of the thermocouples on the PV cells and the experimental setup. The PV module used in this study consisted of 60 monocrystalline passivated emitter and rear cell (PERC) modules linked in parallel and series. The dimension of the PV module was 1000 mm (width) × 1670 mm (height) × 32 mm (depth), and the module had a maximum power of 300 W and open circuit voltage and current of 39.66 V and 9.83 A, respectively. The temperatures of the PV

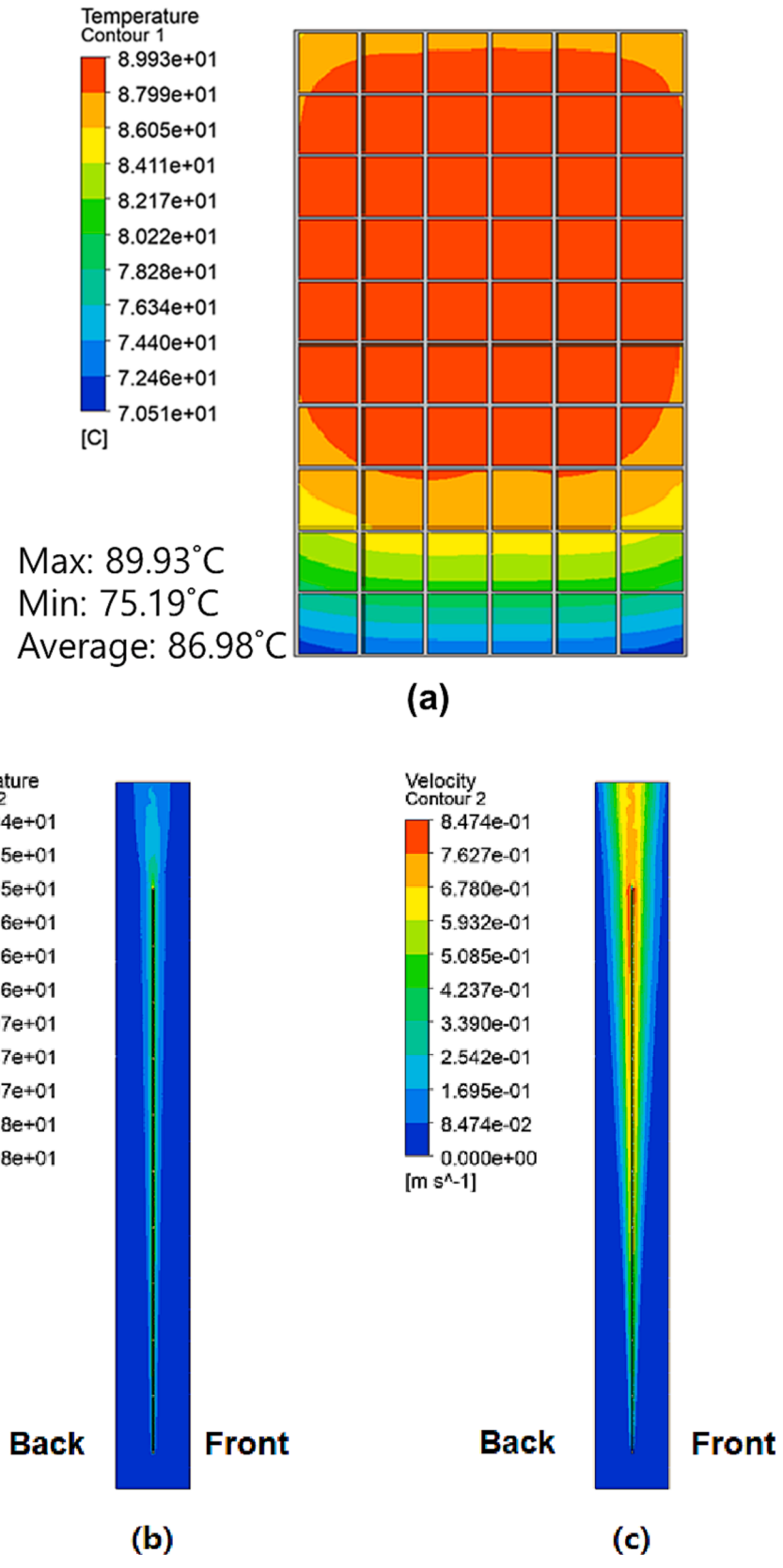


Fig. 7. (a) Temperature distribution of the PV cells in module, and (b) temperature and (c) velocity distribution of the air domain of the module-only model.

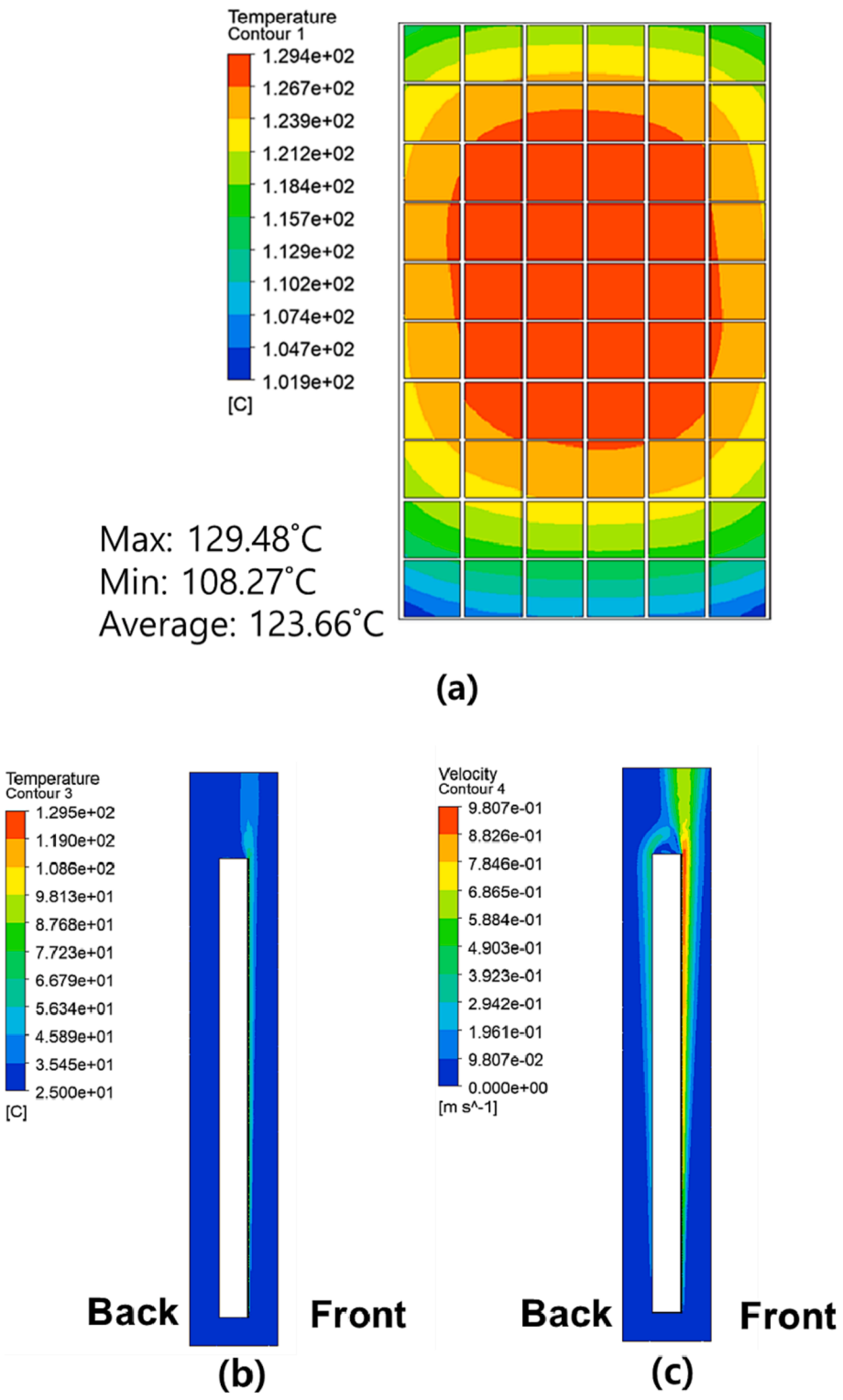


Fig. 8. (a) Temperature distribution of the PV cells in the module, and (b) temperature and (c) velocity distribution of the air domain of the PVNB model without a heat sink.

cells were measured using T-type thermocouples with an accuracy of ± 0.5 ℃ and measuring range of –270 °C to 370 °C. Because the temperature distribution was symmetrical to the centerline, the temperatures of the PV cells on the left side of the PV module were measured using 15 thermocouples attached to the center of each PV cell, as shown in Fig. 4 (a). Temperatures were measured every second using a data acquisition system, and the average values of the temperatures stored for 5 min were used. Because it is difficult to control natural sunlight, and it is affected by various parameters, a solar simulator was manufactured and used instead of natural sunlight radiation. The solar simulator was manufactured by referring to previous studies [28,29]. The solar simulator was equipped with 25 ultraviolet lamps (300 W each) to provide a continuous light spectrum. To measure the intensity of the radiation to the module, a solar pyranometer with an accuracy of ± 3% and

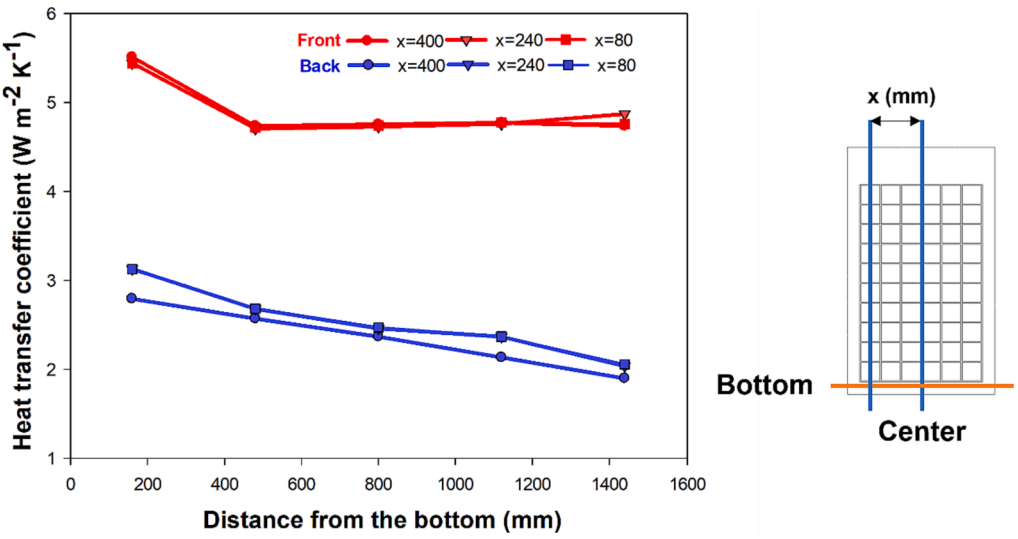


Fig. 9. Convective heat transfer coefficient in the PVNB model without a heat sink at various locations.

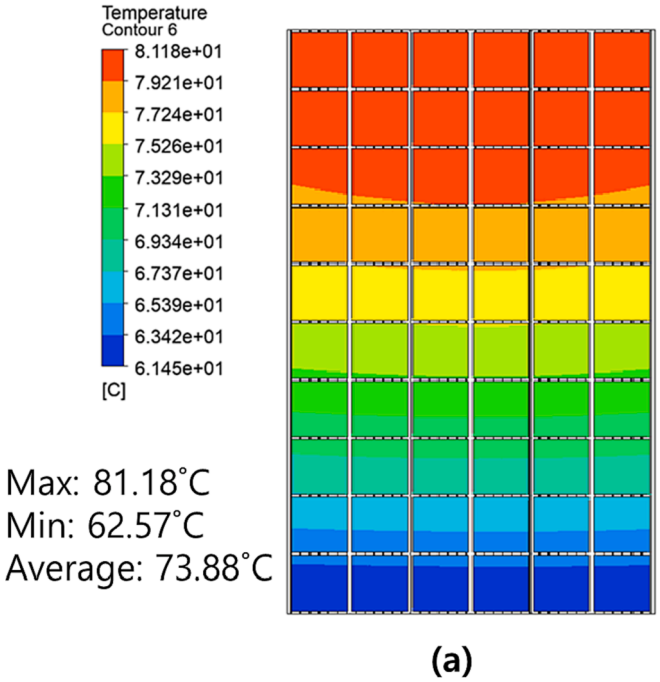


Fig. 10. (a) Temperature distribution of the PV cells, and temperature and velocity distribution of the air domain for (b) overall view and (c) detailed view of the PVNB model with the conventional heat sink.

measuring spectral range of 360–1120 nm was used. The average heat flux of the cells was varied from 300 W m^{-2} to 500 W m^{-2}. Photographs of the experimental equipment are shown in Fig. 5.

The simulation was validated by comparing the predicted and experimental cell temperatures at different heat flux values. As shown in Fig. 6, all the predicted cell temperatures deviated by less than ± 10% from the experimental cell temperature. In addition, the validity of the simulation was verified using the MAE of the cell temperature values, which can be expressed by Eq. (22), showing an MAE of 4.53%.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\frac{\left|T_{measured} - T_{predicted}\right|}{T_{measured}} \times 100 \quad (22)$$

4. Results and discussion

4.1. Effect of the noise barrier

The effect of the noise barrier on the module temperature and cooling performance was analyzed by comparing the simulation results of the module-only and PVNB models without a heat sink. Fig. 7(a) shows the temperature distribution of the PV cells in the module-only model. The maximum module temperature of 89.93 ℃ was found at upper side of the module, and the minimum module temperature of 75.19 ℃ was found at bottom side of the module. This is because the thermal boundary layer is thinner upstream than downstream for natural convection, yielding the highest convective heat transfer coefficient upstream. The average module temperature was 86.98 ℃, which

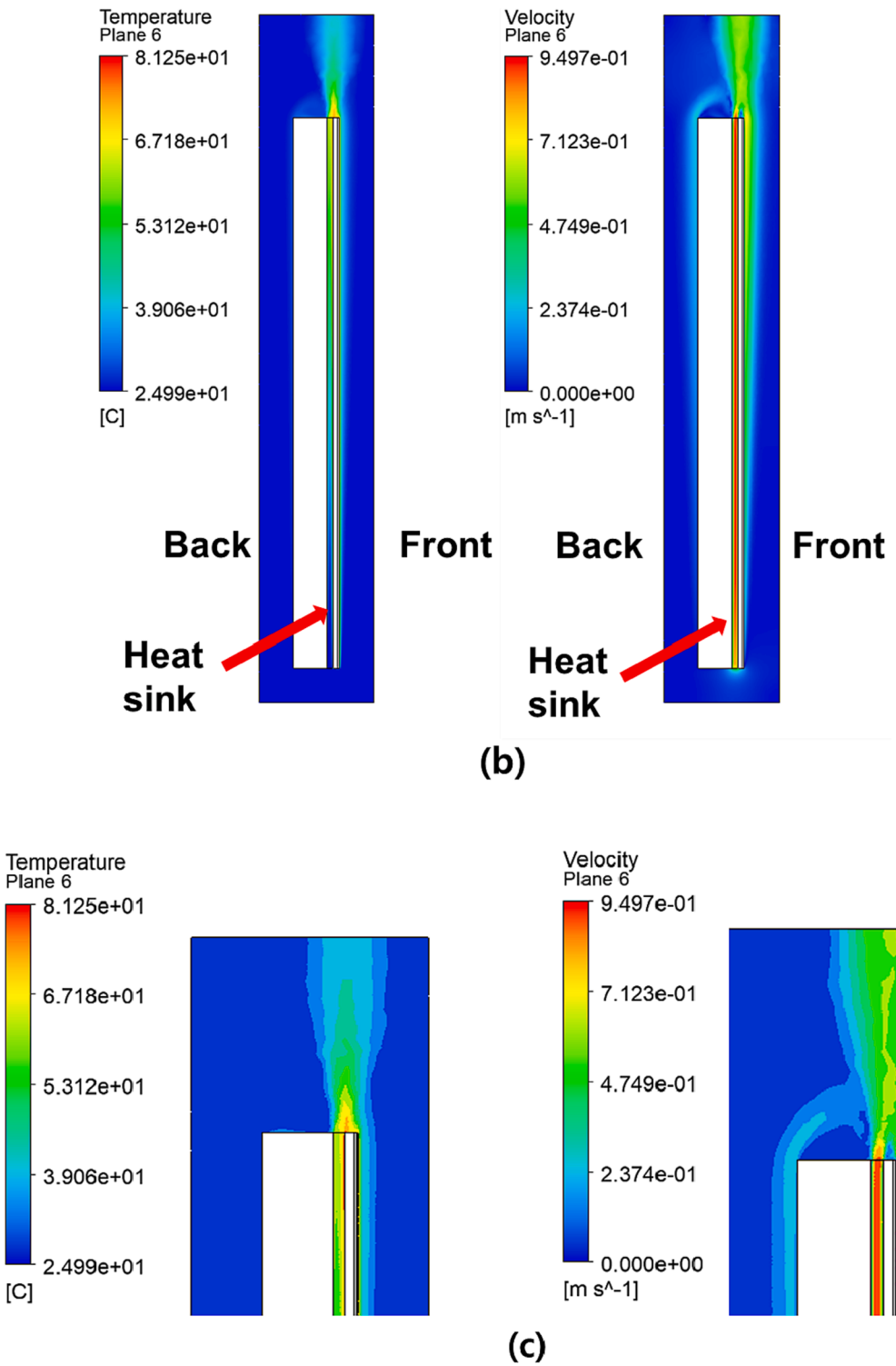


Fig. 10. (*continued*).

showed an acceptable margin of 90 ℃, which is the limit temperature suggested by the manufacturer even with the given strong solar irradiation condition. Fig. 7(b) and 7(c) show the temperature and velocity distributions in the air domain for the module-only model, respectively. As the module was very thin, the temperature difference between the two opposite surfaces of the module was very low. Therefore, the temperature and velocity distributions of air were almost symmetric. The air temperature and velocity gradually increased as the distance from the bottom increased, indicating that heat was transferred from the module to ambient air via natural convection.

Fig. 8 shows the simulation results for the PVNB model without a heat sink. As shown in Fig. 8(a), the temperature distribution showed a similar trend with the module-only model; however, the average temperature of the PV cells for the PVNB model without a heat sink significantly increased by 42.2% from 86.98 ℃ to 123.66 ℃. In addition, the maximum and minimum cell temperature increased by 44.0% from 89.93 ℃ to 129.48 ℃ and by 43.9% from 75.19 ℃ to 108.27 ℃, respectively. This indicates that adding a noise absorber to the backplate

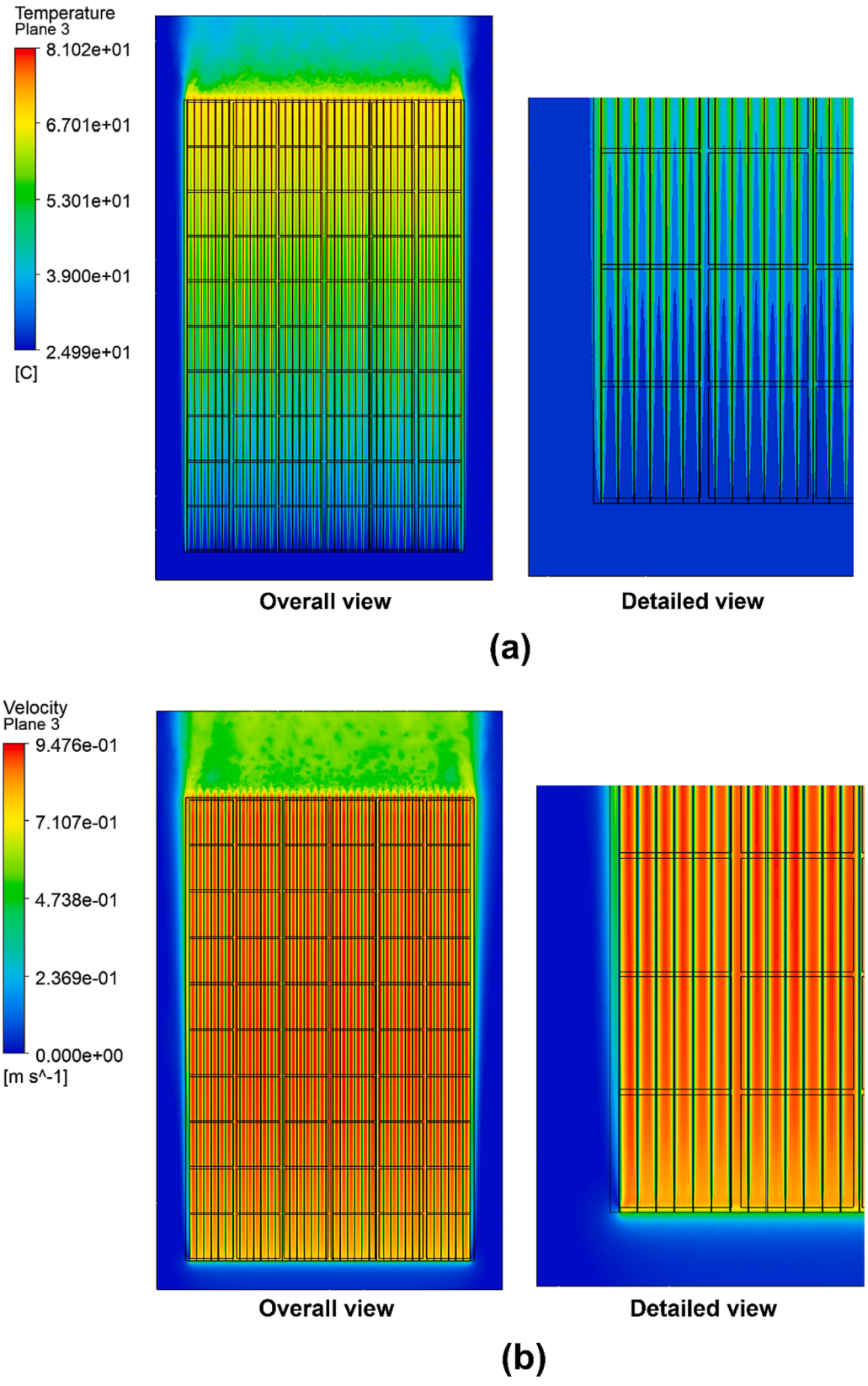


Fig. 11. (a) Temperature and (b) velocity distribution of the air domain for the PVNB model with the conventional heat sink.

of the module significantly decreases the cooling performance of the module owing to the extremely low thermal conductivity of the noise absorber. Fig. 8(b) and 8(c) show the temperature and velocity distributions in the air domain. Unlike in the module-only model, the changes in air temperature and velocity along the flow direction were substantially lower at the back side of the noise absorber owing to the low thermal conductivity of the noise absorber, indicating poor heat transfer by natural convection. That is, the heat of the PV module in the PVNB

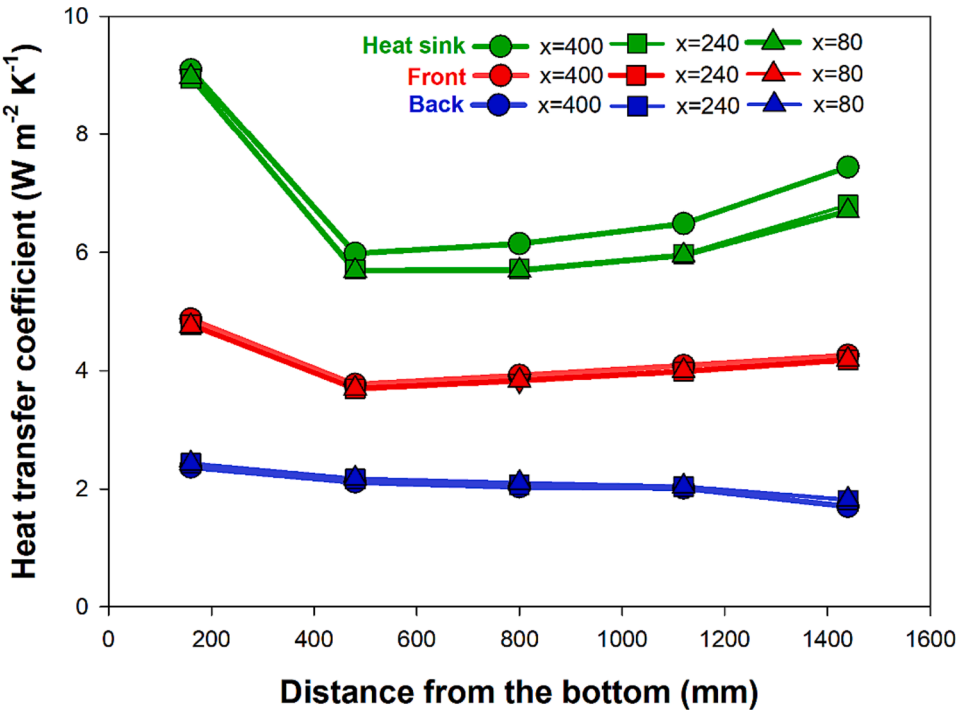


Fig. 12. Heat transfer coefficient of the PVNB model with the conventional heat sink at different locations.

model without a heat sink was mostly transferred to the front side of the PV module, yielding excessive PV cell temperatures.

Fig. 9 shows the convective heat transfer coefficients at various locations in the PVNB model without a heat sink. The heat transfer coefficients of both the front and back sides were the highest at the bottom of the module owing to the thin thermal boundary layer and decreased along the upward flow direction. However, unlike the trend on the back side, the heat transfer coefficient did not show a gradual decrease on the front side because the high temperature of the PV cell accelerated natural convection. The heat transfer coefficients on the front side ranged from 4.71 W m^{-2} ℃$^{-1}$ to 5.51 W m^{-2} ℃$^{-1}$, and those of the back side ranged from 1.90 W m^{-2} ℃$^{-1}$ to 3.13 W m^{-2} ℃$^{-1}$, indicating an approximately doubled heat transfer coefficient for the front side. The difference in the heat transfer coefficient according to the distance from the center of the module was trivial for both sides, owing to a similar temperature distribution. The high heat transfer coefficient on the front side was attributed to the high temperature of the PV module caused by the low cooling performance of the PVNB without a heat sink. In conclusion, a PVNB without a heat sink has dramatically reduced cooling performance, resulting in high PV cell temperatures, which can cause severe degradation in the electrical performance and lifespan of the PV module.

4.2. PVNB with the conventional heat sink

As concluded in the previous section, the PVNB without a heat sink had poor cooling performance; therefore, an additional cooling method needs to be applied to the PVNB. Because PVNBs are generally installed in large-scale constructions such as railroads, highways, and residential areas, practical maintenance and reasonable prices are the most important factors. Therefore, a conventional simple heat sink with a general geometry was used as the passive cooling method, and the effect of using a conventional heat sink was analyzed.

Fig. 10(a) shows the temperature distribution of the PV cells for the PVNB model with a conventional heat sink. Compared with the PVNB model without a heat sink, the average temperature of the PV cells for the PVNB with the conventional heat sink significantly decreased by 40.3% from 123.66 ℃ to 73.88 ℃. Furthermore, the maximum and minimum cell temperatures decreased by 37.3% from 129.48 ℃ to 81.18 ℃ and by 42.2% from 108.264 ℃ to 62.567 ℃, respectively. Fig. 10(b) and 10(c) show the temperature and velocity distributions of air for the PVNB with the conventional heat sink. In contrast to the result of the PVNB without a heat sink, the PV module was cooled by the low-temperature air between the noise absorber and the module because of the air channel created by the heat sink. In particular, the velocity of the air in the heat sink channel was considerably higher than that on the front side of the module, which promoted cooling by natural convection. This is because the buoyancy force applied to the bulk air in the heat sink channel accelerated the air velocity, causing a stack effect. The air trapped in the heat sink channel resulted in a horizontal isothermal contour, as shown in Fig. 10(a), because the air temperature increased along the flow direction.

Fig. 11 shows the temperature and velocity distributions of air in the heat sink channel of the PVNB with a conventional heat sink. Cold air enters the inlet of the heat sink channel from the bottom of the model and exits the top outlet after absorbing heat from the heat sink, as shown in the overall view in Fig. 11(a). For the PVNB with a heat sink, air flowed upward through a square channel created by the heat sink and noise absorber. Therefore, even though the driving force was the buoyancy force owing to the temperature difference of the air, similar to natural convection, the flow showed characteristics of typical internal flow. The detailed view in Fig. 11(a) shows the entrance region of the air in the heat sink channel at the bottom; the two air thermal boundary layers on the opposite side collapsed after a certain length. The temperature of the channel core remained constant at the ambient temperature of 25 ℃ in the entrance region; however, the air temperature at all locations of the heat sink channel increased along the flow direction up to 81 ℃ at the exit, resulting in an increase in the stack effect. Because of the stack effect, the air velocity in the heat sink channel was comparably high up to around 0.95 m s^{-1}, as shown in the overall view in Fig. 11(b). In addition, the air velocity from the inlet of the heat sink channel was high owing to the stack effect, as shown in the detailed view in Fig. 11(b); without the stack effect, the air velocity would have been much lower at the bottom of the model.

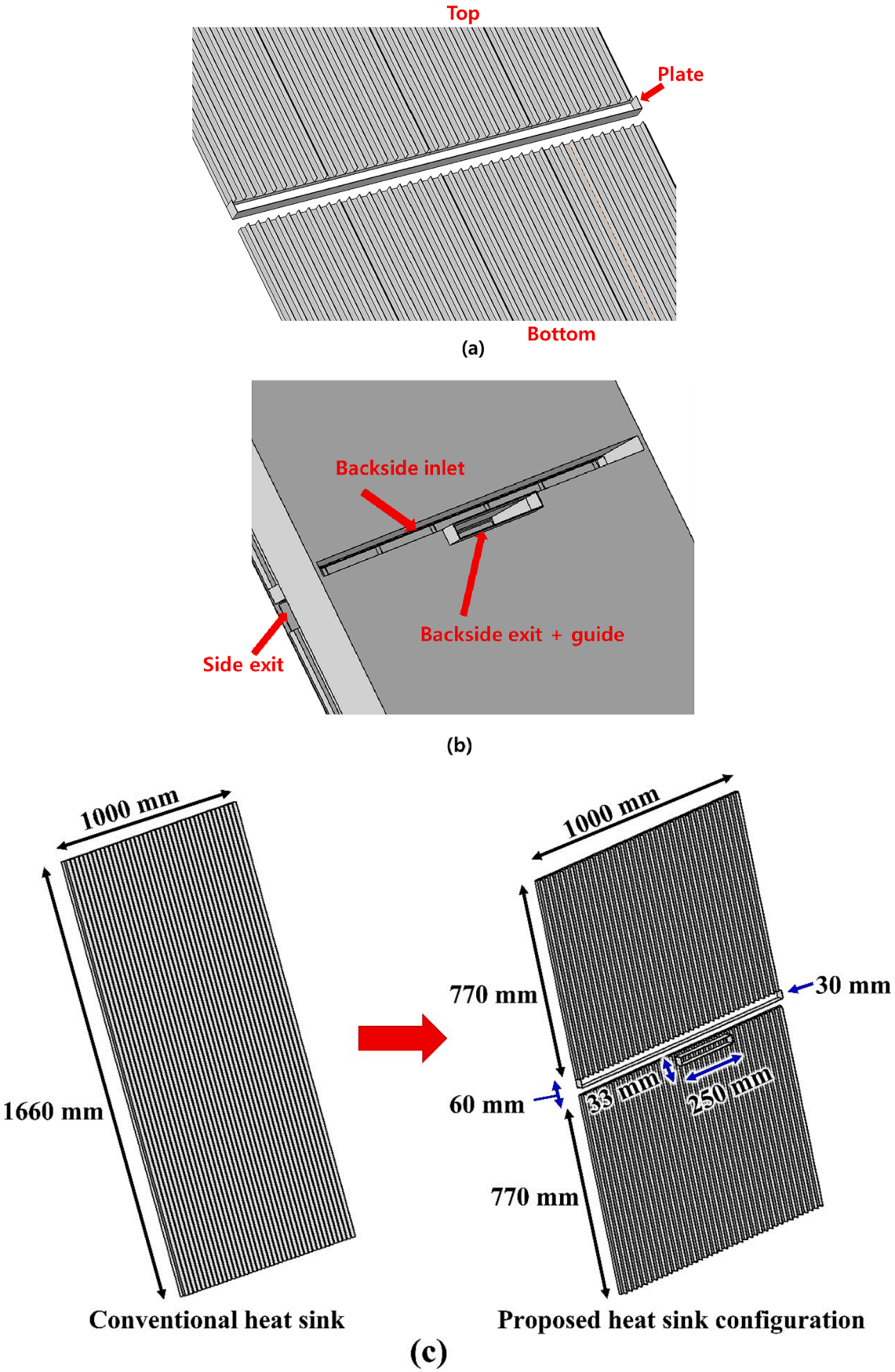


Fig. 13. Geometry of the PVNB model with the proposed heat sink configuration: (a) separation of the top and bottom parts with a plate, (b) addition of the backside inlet and backside exit with a guide at the noise absorber, and (c) major changes and dimensions of the heat sink.

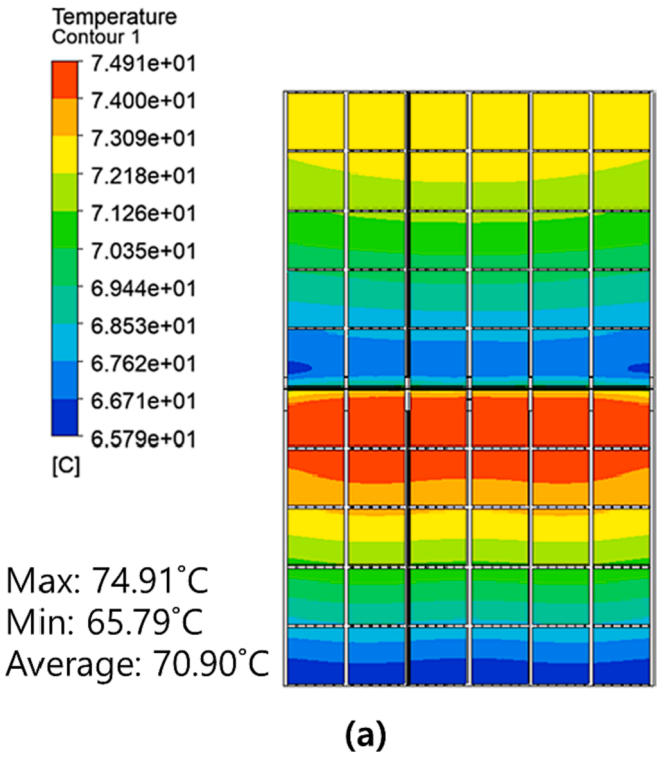


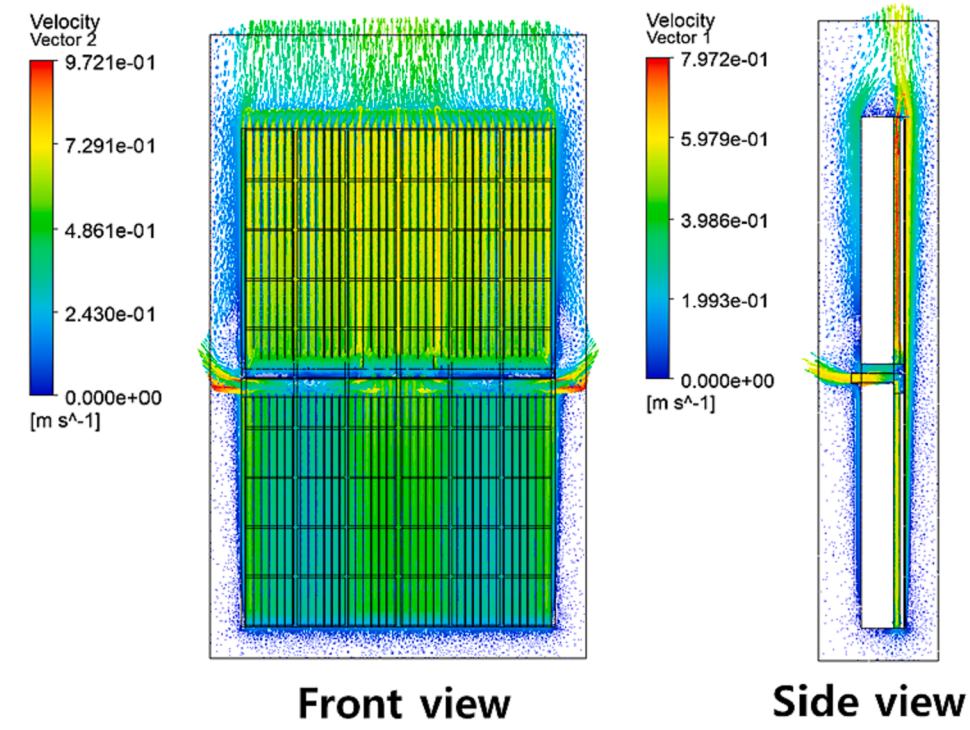
Fig. 14. (a) Temperature distribution of the PV cells, (b) temperature and (c) velocity distribution of the air domain of the proposed model, the detailed view of the air flow in the (d) side exits and the (e) backside inlet and exit at noise absorber.

Fig. 12 depicts the convective heat transfer coefficient of air at various locations in the PVNB model with a conventional heat sink. The trend in the heat transfer coefficients according to the distance from the bottom for both the front and back sides was similar to that of the PVNB model without a heat sink. However, the values of the heat transfer coefficients were lower than those for the PVNB model without a heat sink, ranging 3.70–4.88 W m^{-2} ℃$^{-1}$ and 1.71–2.44 W m^{-2} ℃$^{-1}$ for the front and back side, respectively, owing to the lower surface temperatures because of the better cooling performance. The trends and values of the heat transfer coefficients in the heat sink channel were considerably different from those on the front and back sides. The heat transfer coefficient of air in the heat sink channel was significantly high, up to around 9.01 W m^{-2} ℃$^{-1}$ at the inlet of the channel because of the thin thermal boundary layer at the inlet as well as faster flow velocity than the other two sides owing to the stack effect. Meanwhile, the heat transfer coefficient of air in the heat sink dramatically decreased to approximately 5.80 W m^{-2} ℃$^{-1}$ at a 480 mm distance from the bottom because the two thermal boundary layers at the opposite side of the fin collapsed. Nevertheless, because of the stack effect, the heat transfer coefficients of air in the heat sink channel were still higher than those at the front side, even after a steep drop. After the steep drop, the heat transfer coefficient increased with increasing distance from the bottom owing to the faster flow velocity because the increase in air temperature resulted in an increase in the specific volume, leading to acceleration of the air. The heat transfer coefficients of air in the heat sink channel increased by about 17.75–24.21% from 5.70 to 5.99 W m^{-2} ℃$^{-1}$ to 6.70–7.44 W m^{-2} ℃$^{-1}$ as the distance from the bottom increased from 480 mm to 1440 mm. Therefore, in the PVNB model with the conventional heat sink, the cooling performance was significantly enhanced because of the stacking effect, which increased the flow velocity and heat transfer area owing to the fin structures of the heat sink.

4.3. PVNB with the proposed heat sink configuration for enhancing cooling performance

Although the increase in the air temperature in the heat sink channel increased the stack effect, which enhanced the heat transfer owing to the increased flow velocity, the heated air temperature caused a low cooling performance, especially at the top. Therefore, a novel heat sink configuration with an identical conventional heat sink and better cooling performance was proposed, maintaining the advantages of using a conventional heat sink: practical maintenance and reasonable price. The modifications on the heat sink configuration were conducted without changing the geometry of the heat sink to increase the cooling performance of the PVNB with minimum costs. Fig. 13 shows the geometry of the PVNB model with the proposed heat sink configuration. As shown in Fig. 13(a), in the proposed heat sink configuration, the conventional heat sink was divided into two parts (top and bottom), and a plate was installed between the two parts to separate the heat sink channel. With a plate between the two parts, the heated air in the bottom heat sink channel can be discharged through both the left- and right-side exits before it reaches an excessively high temperature. In addition, as shown in Fig. 13(b), a backside inlet and a backside exit combined with a guide was added at the back of the noise absorber. The backside inlet supplies relatively cold ambient air to the top heat sink channel, and the backside exit discharges hot air in the bottom heat sink channel, together with the side exits at the left and right sides. Without a guide, the heated air from the bottom heat sink channel can re-enter the backside inlet, thereby decreasing the cooling performance of the top heat sink. The air flow area of the PVNB model with the proposed heat sink configuration increased by around only 4% from the PVNB model with the conventional heat sink. Therefore, the proposed model still can benefit from the stack effect also with the increased heat transfer at the middle of the model by the supplied cold ambient air. Fig. 13(c) shows the major changes and the dimensions of the heat sink.

Fig. 14(a) shows the temperature distribution of the PV cells for the



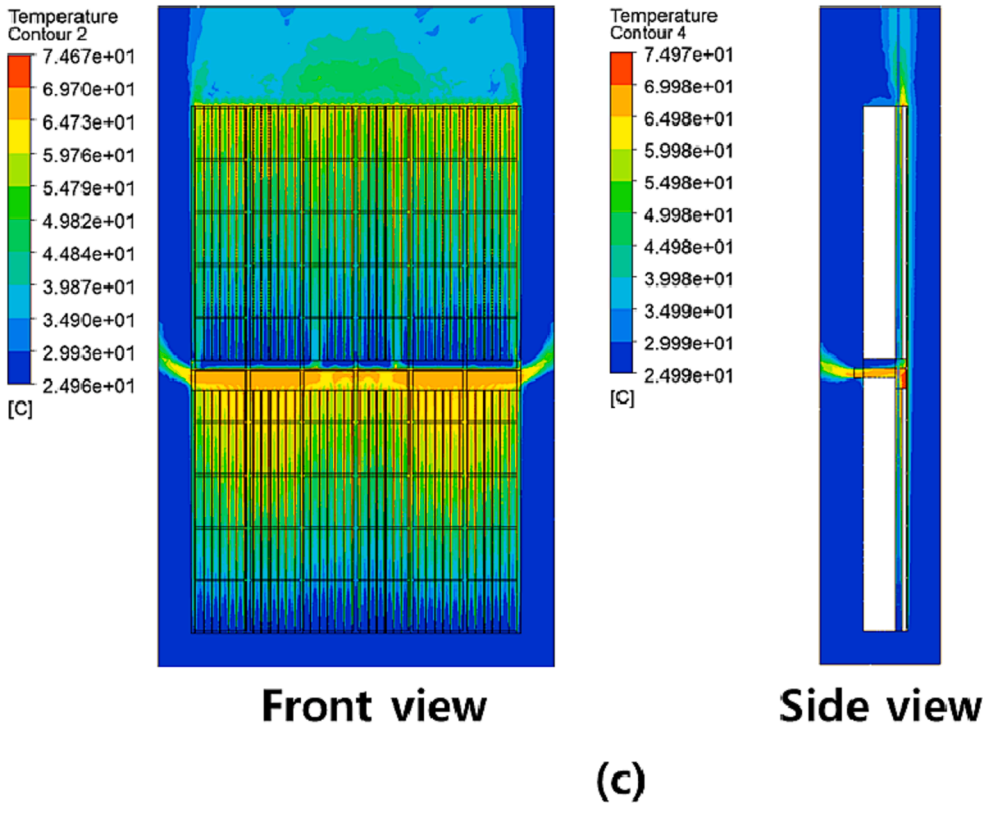
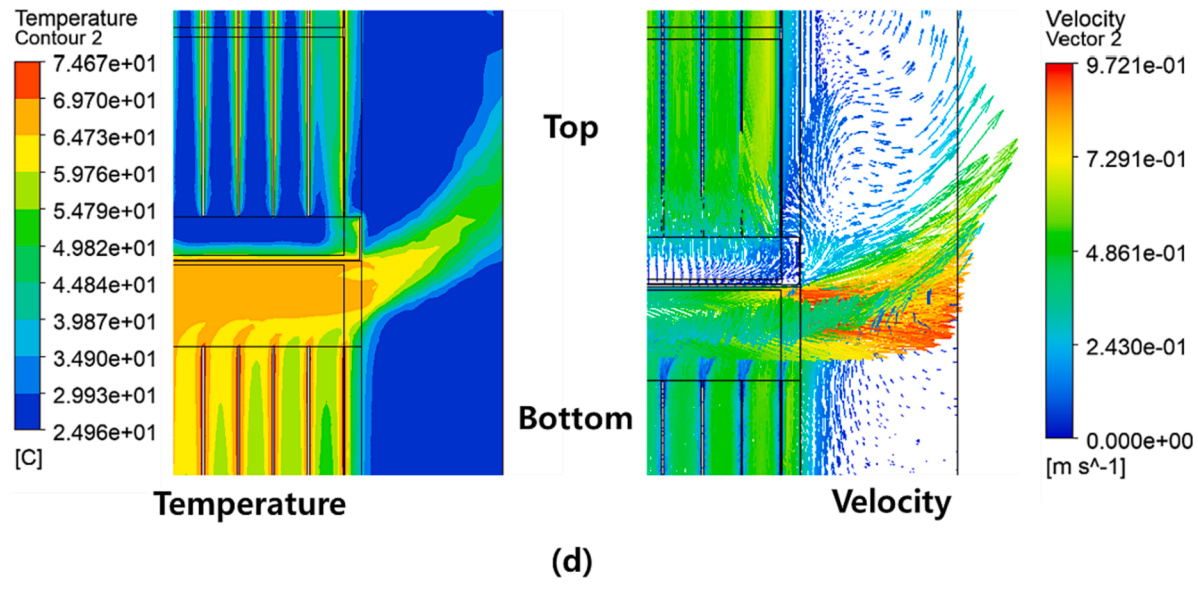


Fig. 14. (*continued*).

PVNB model with the proposed heat-sink configuration. Compared to the previous conventional model, the maximum cell temperature decreased by 7.7% from 81.18 ℃ to 74.91 ℃ and the average cell temperature decreased by 4.1% from 73.88 ℃ to 70.90 ℃, indicating an increase in the cooling performance. However, the minimum cell temperature increased by 5.1% from 62.57 ℃ to 65.79 ℃ because of the decreased air velocity in the heat sink channel caused by the decreased stack effect. These temperature characteristics can be explained by the



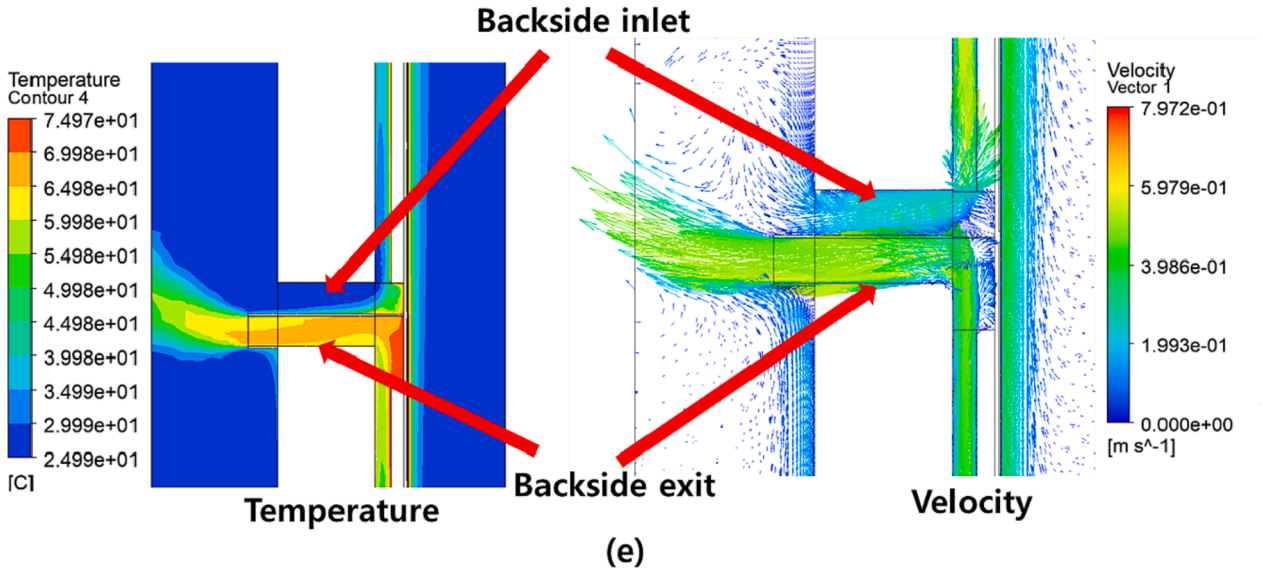


Fig. 14. (*continued*).

airflow characteristics in the heat-sink channel. Fig. 14(b) and 14(c) depict the velocity and temperature distributions of air in the heat sink channel, respectively. As heated air was discharged to the side exits and the backside exit in the noise absorber, the air velocity at the exits was noticeably high, as shown in the detailed view in Fig. 14(d). In addition, the air velocity was higher in the top heat sink channel than in the bottom heat sink channel because of the lack of blockage by the plate at the ceiling of the channel. Because of the blockage by the plate at the ceiling in the bottom heat sink channel, the air temperature was the highest at the top of the bottom heat sink channel. Fig. 14(e) shows the detailed temperature and air distribution near the backside inlet of the noise absorber. The discharged heated air in the bottom heat sink channel and the entering cold ambient air did not mix owing to the guide, indicating that cooling was properly conducted.

Fig. 15 shows the heat transfer coefficient of the air at various locations in the PVNB model with the proposed heat sink configuration, including the front, back, and heat sink channels. Most of the results were similar to those of the previous model with the conventional heat sink because the geometry of the heat sink was nearly identical; however, some deviations were reported owing to the changes in the configuration. The heat transfer coefficients at the front and back sides showed negligible changes compared with those in the previous model; however, the newly developed thermal boundary layer at the top noise barrier increased the heat transfer coefficient and decreased cell temperature owing to the separation of the heat sink. In contrast, the increase in the heat transfer coefficient was significant, especially for the inlet region of the top heat sink channel, because of the cold ambient air supply. The heat transfer coefficient at the inlet of the bottom heat sink channel was lower than that of the top heat sink channel because of the lower flow velocity owing to blockage by the plate at the ceiling. The heat transfer coefficients of air were in the ranges of 6.5–9.6 W m^{-2} $°C^{-1}$ and 7–15 W m^{-2} $°C^{-1}$ for the bottom and top heat sink channels, respectively.

Fig. 16 compares the PV module temperatures and average heat transfer coefficients of the four different simulation models. The PVNB without a heat sink significantly increased the temperatures by 42.2–44.0% compared to the module-only case, mainly because of the decreased heat transfer through the noise absorber. Employing the

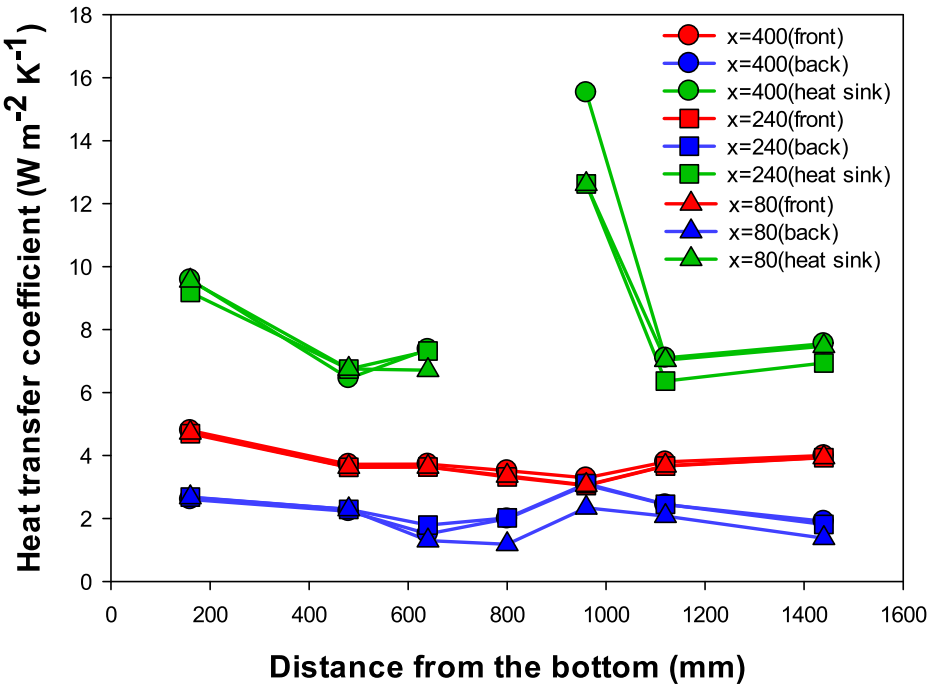


Fig. 15. Heat transfer coefficient of the PVNB model with the proposed heat sink configuration at different locations.

conventional heat sink increased the cooling performance and caused 37.3–42.2% lower temperature than the PVNB without a heat sink, owing to the enhanced heat transfer coefficient of the air flowing in the heat sink channel. The heat transfer coefficient of air in the heat sink channel was 6.76 W m^{-2} ℃$^{-1}$, whereas that at the PV module front was 4.13 W m^{-2} ℃$^{-1}$. The proposed heat sink configuration further decreased the PV module temperatures by 39.3–42.7% compared with the PVNB without a heat sink, owing to the improved air flow with a 25.6% increase in the heat transfer coefficient in the heat sink channel compared with the conventional heat sink case.

4.4. Electrical performance analysis

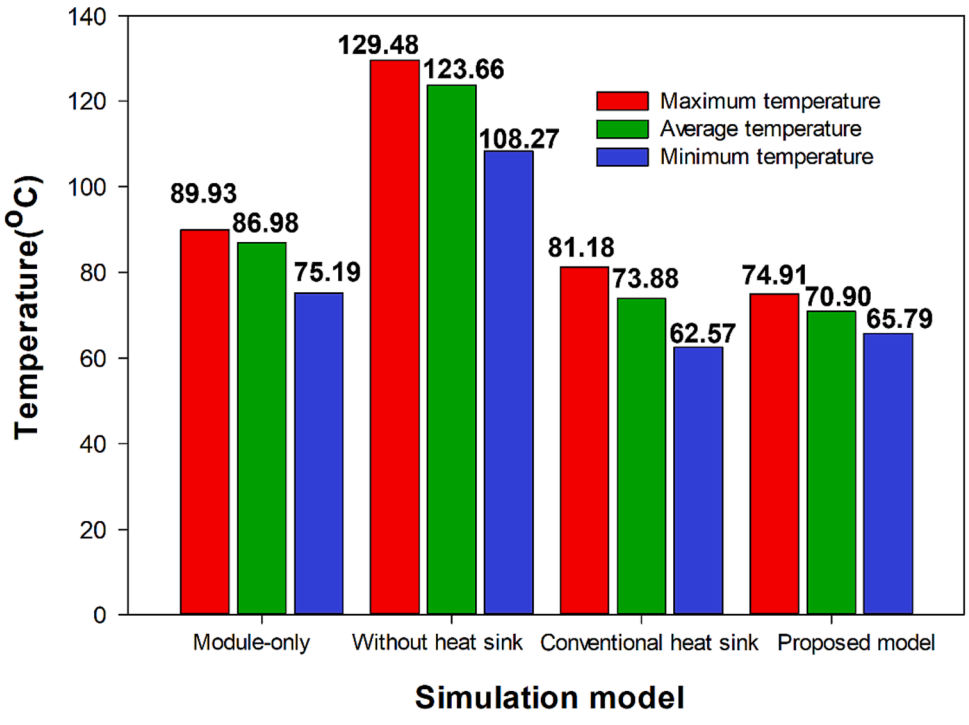
The electrical performance of the PV modules in the investigated PVNB models was evaluated by calculating the output power and open-circuit voltage. Fig. 17 shows the P–V curve and I–V curves of the PV modules for 4 different PVNB models. All curves showed a similar trend, except for the maximum power and open-circuit voltage. The open-circuit voltage and maximum output power increased as the average module temperature decreased. This is because the band gap of the semiconductor decreases as the temperature increases, resulting in a decrease in the generated energy [30].

Fig. 18 shows the maximum power and open-circuit voltage of the PV modules for the studied PVNB models. Compared to the module-only model, the PVNB model without a heat sink significantly decreased the maximum output power by 15.5% from 204.54 W to 172.80 W and the maximum voltage by 11.8% from 32.71 V to 28.90 V, owing to the low thermal conductivity of the noise absorber. In the case of the PVNB model with the conventional heat sink, the maximum output power and voltage increased to 216.14 W and 34.03 V, respectively, which were 25.1% and 17.8% higher than those of the PVNB model without a heat sink; these values were 5.68% and 4.07% higher than the module-only model, respectively. After some simple modifications in the configuration of the conventional heat sink, the maximum output power and voltage increased by 26.6% to 218.80 W and by 19.0% to 34.30 V, respectively, compared to the PVNB model without a heat sink. The efficiency improvement of the PV module in the PVNB with the proposed heat sink configuration is significant, and the increase in annual power generation should be considerable because the PV module operates over a year for its entire lifespan of around 30–35 years.

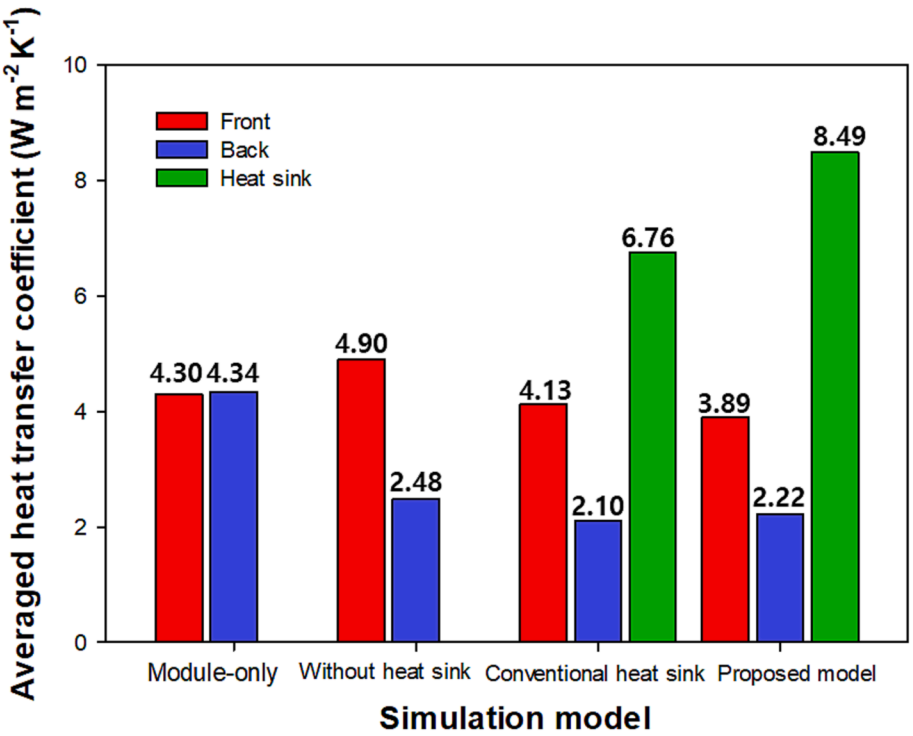
5. Conclusions

In this study, the cooling and electrical performances of a PVNB were investigated using four different models: a PV module-only model, a PVNB model without a heat sink, a PVNB model with a conventional heat sink, and a PVNB model with the proposed heat sink configuration. The key findings and investigations in this study are as follows:

- The average cell temperature in the PVNB model without a heat sink significantly increased by 42.2% from 86.98 ℃ to 123.66 ℃ owing to the low thermal conductivity of the noise absorber when compared to the module-only model.
- The average cell temperature in the PVNB with the convectional heat sink significantly decreased by 40.3% from 123.66 ℃ to 73.88 ℃ when compared with the PVNB model without a heat sink. This was because the PV module was cooled by the low-temperature air in the heat sink channel, and heat transfer was promoted by the stack effect. However, the cooling performance at the top was poor because of the high air temperature.
- A novel heat sink configuration with improved cooling performance was proposed, and the average cell temperature decreased by 42.7% from 123.66 ℃ to 70.90 ℃ owing to the additional supply of the cold ambient temperature air to the heat sink channel when compared to the PVNB model without a heat sink.
- The maximum output power and voltage increased by 25.1% and 17.8% for the PVNB model with the conventional heat sink, respectively, and increased by 26.6% and 19.0% for the PVNB with the proposed heat sink configuration, respectively, compared with the PVNB model without a heat sink. This was because the open-circuit voltage and maximum output power increased as the average cell temperature decreased owing to an increase in the band gap.



(a)



(b)

Fig. 16. Comparison of the (a) PV module temperatures and the (b) averaged heat transfer coefficients for the simulation models.

CRediT authorship contribution statement

Hyun Sung Hwang: Methodology, Investigation, Writing – original draft, Visualization. **Sungho Yun:** Formal analysis, Validation. **Jae-Chul Kim:** Data curation, Conceptualization. **Dongchan Lee:** Supervision, Writing – review & editing, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial

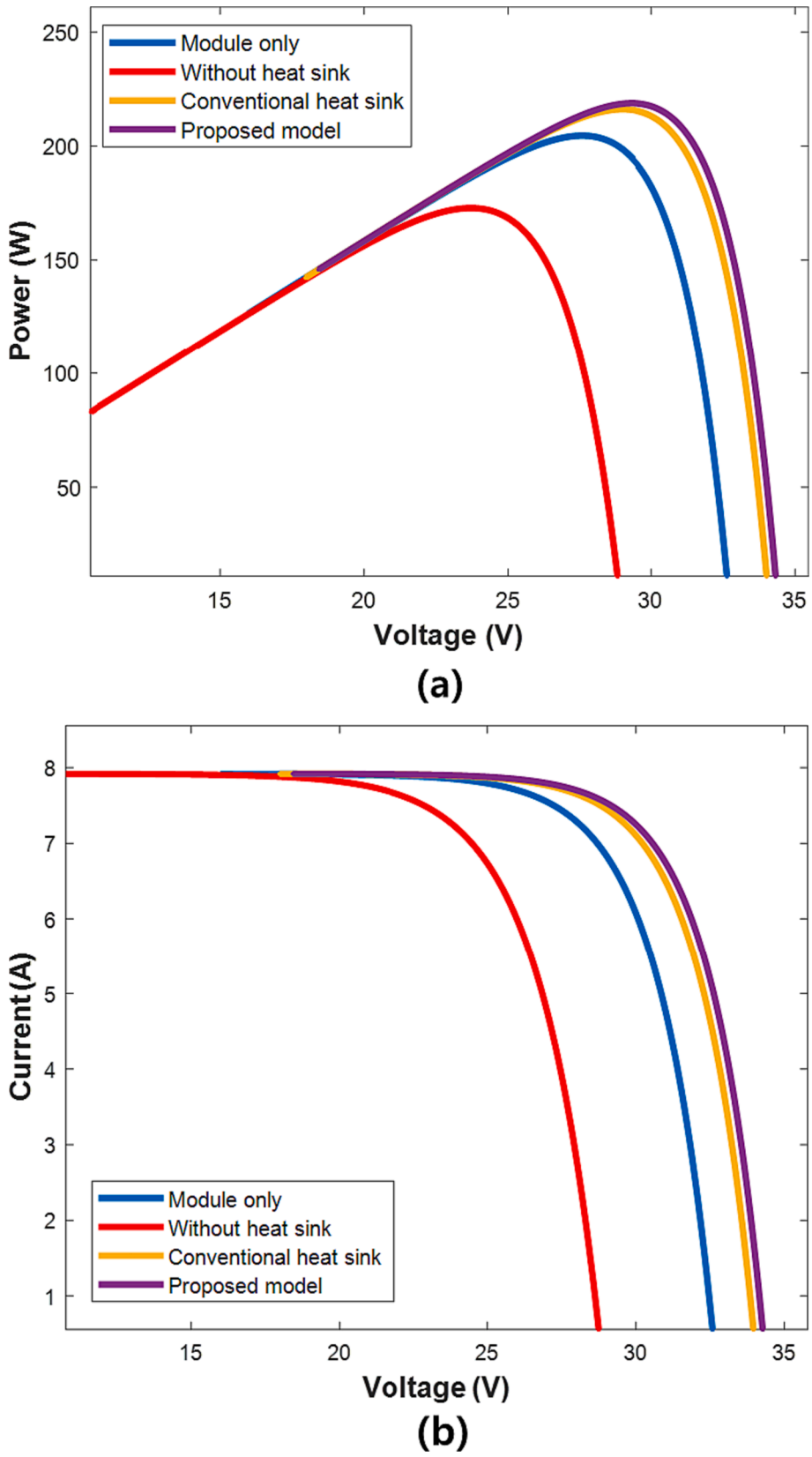


Fig. 17. (a) P–V curves and (b) I–V curves of the PV modules for the studied PVNB models.

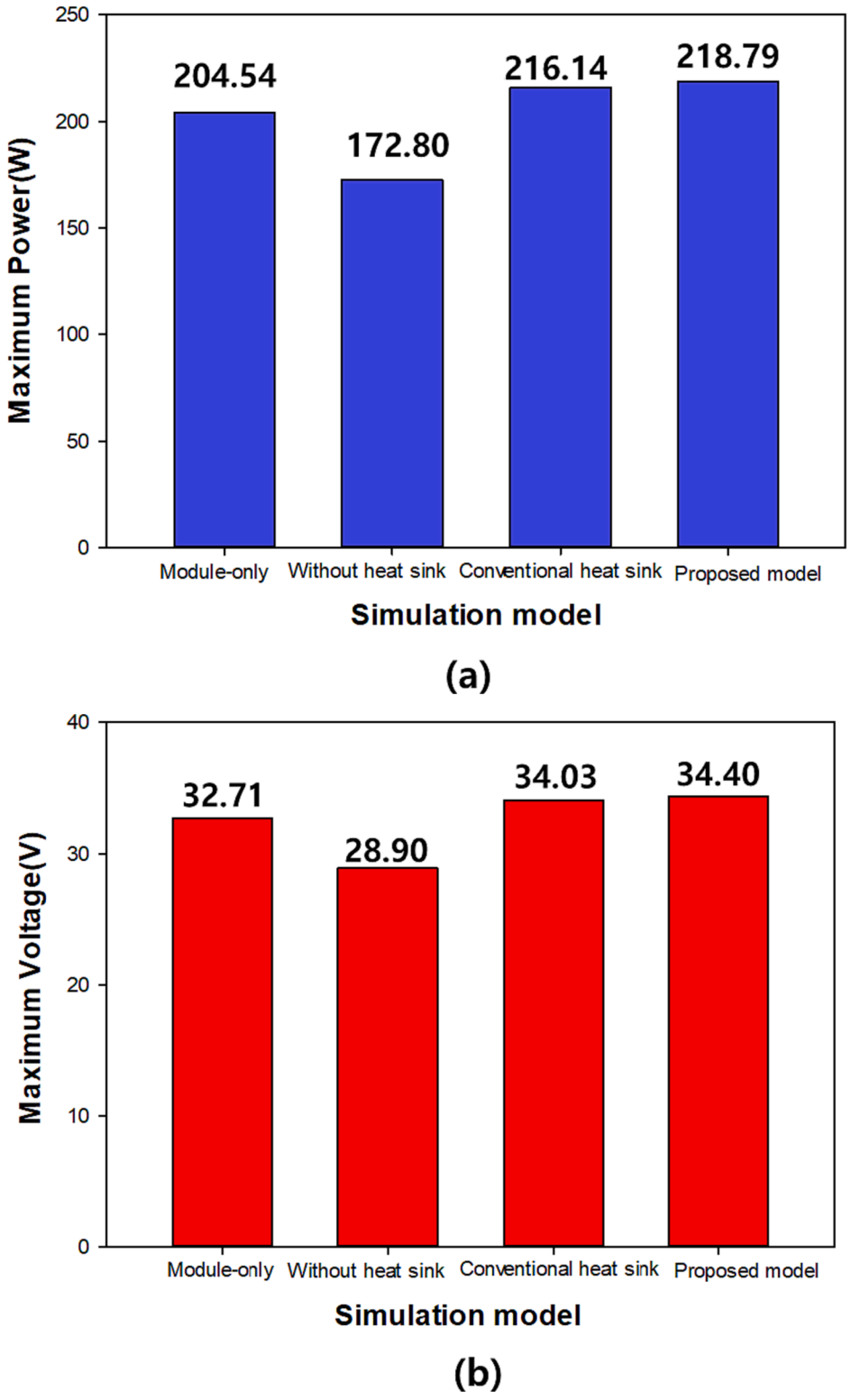


Fig. 18. Comparison of the (a) maximum output power and the (b) maximum voltage of the PV modules for the studied PVNB models.

interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors are unable or have chosen not to specify which data has been used.

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